

Lecture 1: Study Design and Descriptive studies

1. Studies are classified according to the presence of intervention in the exposure into:
 - a. Interventional studies
 - b. Observational studies:
 - i. Have comparison group: analytical
 - ii. No comparison group: Descriptive:
 1. On population level: Ecological studies
 2. On individual level: case report, case series, survey and cross-sectional studies
2. Descriptive studies:
 - a. Observational (without intervention) studies with no comparison group
 - b. Utilities (strengths):
 - i. Describe the disease (who is affected, where and when)
 - ii. Estimating Prevalence (frequency)
 - iii. Generate hypotheses
 - c. Limitations: cannot test hypothesis
3. Ecological studies:
 - a. Descriptive studies (no intervention, comparison group)
 - b. Done on population level
 - c. The unit of observation is the whole population
 - d. Average data for variables are obtained for the whole population to know the type of relationship between them:
 - i. Positive correlation

- ii. Negative correlation
 - iii. No correlation
- e. Limitations:
 - i. Ecological fallacy: use of average data does not ensure exposures and outcomes are in the same individual (exposure and outcomes in the same population not necessarily same person so no correlation on individual level)
 - ii. Cannot control confounding factors and to be confounding factor it must:
 - 1. Be associated with exposure
 - 2. Can cause the outcome
 - 3. Does not lie in a chain of sequence between the exposure and the outcome
 - iii. Data could be missed or inadequate
 - iv. Cannot test hypothesis (descriptive study)
- f. Strengths:
 - i. Relays on already available data
 - ii. Because they are descriptive study:
 - 1. Easy, cheap and quick
 - 2. Generate hypothesis (not testing)
- 4. Individual based studies: individuals are the observational units:
 - a. Case report:
 - i. Single interesting or unusual case
 - ii. Strengths:
 - 1. Because they are descriptive study:
 - a. Quick and cheap
 - b. Highlight unexpected findings so generate hypothesis

iii. Limitations:

1. Data is for single case so cannot be generalized
2. Cannot test hypothesis (as they are descriptive study)

b. Case series:

- i. Group of people but same unusual conditions (multiple but similar case reports)

ii. Strengths:

1. Generate (suggest) new hypothesis (also disease and epidemics) as they are descriptive study
2. Gather more attention so trigger large scale investigations
3. Individuals are included in further studies (case control)

iii. Limitations:

1. Cannot test hypothesis by itself (descriptive study with no comparison group)